

Seeing Data: critique pack

Artefact

Use assets/program_satisfaction_truncated.png.

The chart compares satisfaction after two delivery formats:

- Standard format: 82%
- Redesigned format: 86%

The vertical axis begins at 80%, and the title reads “Redesign dramatically improves satisfaction.” The values are genuine measurements, but no sampling method, group sizes, uncertainty or assignment process is supplied.

Generated interpretation

The redesigned format caused a dramatic improvement in student satisfaction. The four-point increase proves that the new approach is substantially more effective and should replace the standard format across all subjects.

Student task

Annotate each sentence or clause as:

- supported directly;
- plausible but unverified;
- unsupported;
- contradicted by the available evidence.

Then complete the Critique and Repair activity sheet.

Verification questions

- Were participants randomly assigned?
- How many responses were collected in each group?
- Are the groups comparable?
- What were the confidence intervals?
- Was the same survey instrument used?
- Is a four-percentage-point difference practically important?
- Does the result generalise to other subjects?

Repair target

Use assets/program_satisfaction_repaired.png only after students have proposed their own repair.

Suggested caption:

Reported satisfaction was 86% for the redesigned format and 82% for the standard format, a four-percentage-point difference. Without sample sizes, uncertainty or information about group assignment, the comparison does not establish that the redesign caused the difference.